

MACHINE LEARNING PROJECT: PREDICTING SOCIAL MEDIA PLATFORM FROM TREND DATA

MACHINE LEARNING MODEL: LOGISTIC REGRESSION

LOGISTIC REGRESSION : A SUPERVISED MACHINE LEARNING ALGORITHM USED FOR BINARY CLASSIFICATION TASKS, IT PREDICTS WHICH OF TWO CATEGORIES A DATA POINT BELONGS TO. IT DOES THIS BY MODELING THE RELATIONSHIP BETWEEN INDEPENDENT VARIABLES AND THE PROBABILITY OF BELONGING TO A SPECIFIC CLASS, USING A SIGMOID FUNCTION TO MAP THE OUTPUT TO A PROBABILITY BETWEEN 0 AND 1.

- THE DATASET IS Viral_Social_media_Trends.csv
- THIS DATASET CONTAINS 8 FEATURES

STEPS:

1. IMPORTING THE LIBRARY FILES
2. READING THE DATASET
3. PREPROCESSING
4. SPLIT THE DATASET INTO TRAINING AND TESTING
5. BUILD THE MODEL
6. EVALUATE THE PERFORMANCE OF THE MODEL
7. VISUALIZE THE MODEL

1). IMPORTING THE LIBRARY FILES

```
In [ ]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn import linear_model
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix
from sklearn import metrics
```

IMPORT THE DRIVE MODULE FROM THE GOOGLE.COLAB PACKAGE

```
In [ ]: from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

2). READING THE DATASET

```
In [ ]: df = pd.read_csv("/content/drive/My Drive/Viral_Social_Media_Trends.csv")
```

```
In [ ]: df = df.drop('Post_ID', axis=1)
```

PRINT THE SHAPE(DIMENSIONS) OF DATAFRAME

```
In [ ]: print(df.shape)
```

(5000, 9)

PREVIEW THE TOP ROWS OF THE DATASET

```
In [ ]: print(df.head())
```

	Platform	Hashtag	Content_Type	Region	Views	Likes	Shares	\
0	TikTok	#Challenge	Video	UK	4163464	339431	53135	
1	Instagram	#Education	Shorts	India	4155940	215240	65860	
2	Twitter	#Challenge	Video	Brazil	3666211	327143	39423	
3	YouTube	#Education	Shorts	Australia	917951	127125	11687	
4	TikTok	#Dance	Post	Brazil	64866	171361	69581	

	Comments	Engagement_Level
0	19346	High
1	27239	Medium
2	36223	Medium
3	36806	Low
4	6376	Medium

FEATURES OF THE DATASET

```
In [ ]: x = df[['Hashtag', 'Content_Type', 'Region']]
```

CLASS LABEL OF THE DATASET

```
In [ ]: y = df['Platform']
y
```

```
Out[ ]: Platform
0      TikTok
1  Instagram
2      Twitter
3    YouTube
4      TikTok
...      ...
4995  YouTube
4996  Instagram
4997  YouTube
4998  Instagram
4999  Instagram
```

5000 rows × 1 columns

dtype: object

3). PREPROCESSING

```
In [ ]: # Convert categorical columns to numeric using one-hot encoding
X_encoded = pd.get_dummies(x, drop_first=True)

# Now apply StandardScaler
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_scaled = sc.fit_transform(X_encoded)
X_scaled
```

```
Out[ ]: array([[ -0.33518212, -0.33184987, -0.34251779, ..., -0.36647138,
         2.59383409, -0.3957325 ],
        [ -0.33518212, -0.33184987,  2.91955639, ..., -0.36647138,
        -0.38552967, -0.3957325 ],
        [ -0.33518212, -0.33184987, -0.34251779, ..., -0.36647138,
        -0.38552967, -0.3957325 ],
        ...,
        [ -0.33518212, -0.33184987, -0.34251779, ..., -0.36647138,
        -0.38552967,  2.52695951],
        [ -0.33518212, -0.33184987, -0.34251779, ..., -0.36647138,
        -0.38552967, -0.3957325 ],
        [ -0.33518212, -0.33184987, -0.34251779, ...,  2.72872607,
        -0.38552967, -0.3957325 ]])
```

4). SPLIT THE DATASET INTO TRAINING & TESTING

```
In [ ]: from sklearn.model_selection import train_test_split
```

```
xtrain, xtest, y_train, y_test = train_test_split(X_encoded, y, test_size=0.25, random_state = 16)
```

5). **BUILD THE MODEL(LOGISTIC REGRESSION)**

```
In [ ]: from sklearn.linear_model import LogisticRegression
# instantiate the model (using the default parameters)
logreg = LogisticRegression(random_state=16)
# fit the model with data
logreg.fit(xtrain, y_train)
y_pred = logreg.predict(xtest)
```

Confusion Matrix Calculation

```
In [ ]: from sklearn import metrics
cnf_matrix = metrics.confusion_matrix(y_test, y_pred)
cnf_matrix
```

```
Out[ ]: array([[ 91,  52,  64,  71],
               [113,  77,  66,  82],
               [ 74,  72,  55,  83],
               [102,  88,  75,  85]])
```

6). **EVALUATE THE PERFORMANCE OF THE MODEL**

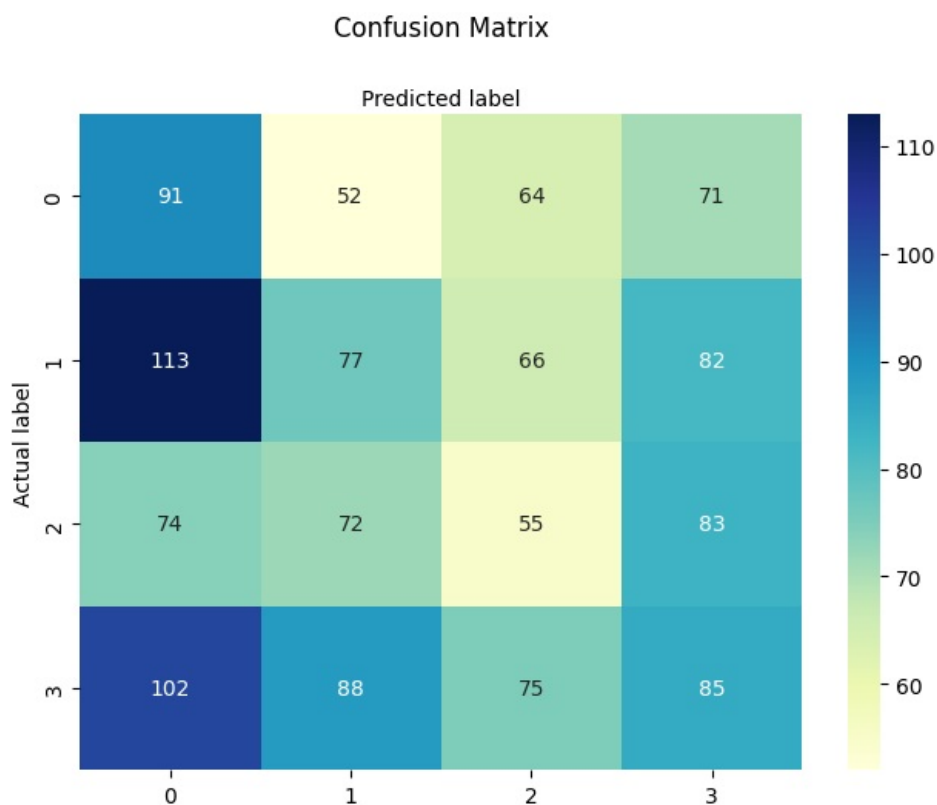
```
In [ ]: # Step 1: Compute the confusion matrix
cnf_matrix = confusion_matrix(y_test, y_pred)
class_names = [0, 1] # Assuming binary classification

# Step 2: Plot the heatmap
fig, ax = plt.subplots()
tick_marks = np.arange(len(class_names))

# Axis ticks
plt.xticks(tick_marks, class_names)
plt.yticks(tick_marks, class_names)

# Heatmap
sns.heatmap(pd.DataFrame(cnf_matrix), annot=True, cmap="YlGnBu", fmt='g')

# Plot styling
ax.xaxis.set_label_position("top")
plt.tight_layout()
plt.title('Confusion Matrix', y=1.1)
plt.ylabel('Actual label')
plt.xlabel('Predicted label')
plt.show()
```



CLASSIFICATION REPORT

```
In [ ]: from sklearn.metrics import classification_report
target_names = y.unique()
print(classification_report(y_test, y_pred, target_names=target_names))
```

	precision	recall	f1-score	support
TikTok	0.24	0.33	0.28	278
Instagram	0.27	0.23	0.25	338
Twitter	0.21	0.19	0.20	284
YouTube	0.26	0.24	0.25	350
accuracy			0.25	1250
macro avg	0.25	0.25	0.24	1250
weighted avg	0.25	0.25	0.24	1250

```
In [ ]: print('Accuracy of the classifier is %0.2f' % metrics.accuracy_score(y_test,y_pred))
```

Accuracy of the classifier is 0.25

MACHINE LEARNING MODEL: SUPPORT VECTOR MACHINE

SVM : A SUPPORT VECTOR MACHINE (SVM) IS A SUPERVISED MACHINE LEARNING ALGORITHM USED FOR BOTH CLASSIFICATION AND REGRESSION TASKS. ITS CORE PRINCIPLE INVOLVES FINDING A HYPERPLANE THAT OPTIMALLY SEPARATES DATA POINTS INTO DIFFERENT CLASSES WHILE MAXIMIZING THE MARGIN BETWEEN THE HYPERPLANE AND THE NEAREST DATA POINTS.

IMPORTING THE LIBRARY FILES

```
In [ ]: from sklearn.svm import SVC
```

SPLIT THE DATASET INTO TRAINING & TESTING

```
In [ ]: xtrain, xtest, ytrain, ytest = train_test_split(x, y, test_size=0.30)
```

BUILD THE MODEL(SVM)

```
In [ ]: from sklearn.preprocessing import LabelEncoder
# Assuming 'xtrain' is a pandas DataFrame and has categorical columns
label_encoder = LabelEncoder()

# Apply label encoding to all columns that contain non-numeric data
xtrain_encoded = xtrain.apply(lambda col: label_encoder.fit_transform(col) if col.dtype == 'object' else col)

# Initialize the SVC model
svc_model = SVC() # You can customize this with parameters if needed

# Now fit the SVC model
svc_model.fit(xtrain_encoded, ytrain)
```

```
Out[ ]: SVC
SVC()
```

DATA PREPROCESSING FOR MODEL PREDICTION

```
In [ ]: from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer

# Assuming your categorical columns are of type 'object'
categorical_columns = xtrain.select_dtypes(include=['object']).columns

# This will transform categorical columns into one-hot encoding
column_transformer = ColumnTransformer(
    transformers=[
        ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_columns)],
```

```

    remainder='passthrough')
# Create a pipeline to preprocess and fit the SVC model
pipeline = Pipeline(steps=[
    ('preprocessor', column_transformer), # Apply the transformations
    ('svc', SVC()) # Apply SVC model after transformation
])
# Fit the model with the pipeline (this will automatically apply transformations)
pipeline.fit(xtrain, ytrain)
ypred = pipeline.predict(xtest)

```

```

In [ ]: i = 0
print ("\n-----")
print ('%-25s %-25s %-25s' % ('Original Label', 'Predicted Label', 'Correct/Wrong'))
print ("-----")
for label in ytest:
    print ('%-25s %-25s' % (label, ypred[i]), end=" ")
    if (label == ypred[i]):
        print (' %-25s' % ('Correct'))
    else:
        print (' %-25s' % ('Wrong'))
    i = i + 1
print ("-----")

```

Original Label	Predicted Label	Correct/Wrong
TikTok	YouTube	Wrong
Instagram	YouTube	Wrong
Instagram	YouTube	Wrong
YouTube	YouTube	Correct
TikTok	Twitter	Wrong
Instagram	Instagram	Correct
YouTube	TikTok	Wrong
TikTok	TikTok	Correct
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```
In [ ]: print("\nClassification Report:\n",metrics.classification_report(ytest, ypred))
print ("-----")
```

```
Classification Report:
              precision    recall  f1-score   support

   Instagram      0.23      0.25      0.24      357
     TikTok      0.23      0.18      0.20      397
     Twitter      0.21      0.21      0.21      353
     YouTube      0.20      0.23      0.21      393

   accuracy              0.22      1500
  macro avg      0.22      0.22      0.22      1500
weighted avg      0.22      0.22      0.22      1500
```

```
In [ ]: print('Accuracy of the classifier is %0.2f' % metrics.accuracy_score(ytest,ypred))
print ("-----")
```

```
Accuracy of the classifier is 0.22
-----
```

MACHINE LEARNING MODEL: K-NEAREST NEIGHBOR

K-NEAREST NEIGHBOR : THE K-NEAREST NEIGHBORS (KNN) ALGORITHM IS A SIMPLE YET POWERFUL MACHINE LEARNING TECHNIQUE USED FOR BOTH CLASSIFICATION AND REGRESSION TASKS. IT PREDICTS THE CLASS OR VALUE OF A NEW DATA POINT BY LOOKING AT ITS "K" NEAREST NEIGHBORS IN THE TRAINING DATA. THE ALGORITHM CALCULATES DISTANCES BETWEEN THE NEW DATA POINT AND ALL POINTS IN THE TRAINING SET, SELECTS THE K CLOSEST NEIGHBORS, AND THEN EITHER ASSIGNS THE MOST FREQUENT CLASS LABEL

IMPORTING THE LIBRARY FILES

```
In [ ]: from sklearn.neighbors import KNeighborsClassifier
```

SPLIT THE DATASET INTO TRAINING & TESTING

```
In [ ]: xtrain, xtest, ytrain, ytest = train_test_split(x, y, test_size=0.30)
```

BUILD THE MODEL(K-NN)

```
In [ ]: # One-hot encode both training and test sets
Xtrain_encoded = pd.get_dummies(xtrain)
xtest_encoded = pd.get_dummies(xtest)

# Align test set to match train set's columns (fill missing columns with 0)
xtest_encoded = xtest_encoded.reindex(columns=Xtrain_encoded.columns, fill_value=0)
```

FIT THE MODEL AGAIN AND PREDICT

```
In [ ]: knn = KNeighborsClassifier(n_neighbors=5)
classifier = knn.fit(Xtrain_encoded, ytrain)
ypred = classifier.predict(xtest_encoded)
```

MODEL EVALUATION - RESULT COMPARISON

```
In [ ]: i = 0
print("\n-----")
print('%-25s %-25s %-25s' % ('Original Label', 'Predicted Label', 'Correct/Worng'))
print("-----")
for label in ytest:
    print('%-25s %-25s' % (label, ypred[i]), end="")
    if(label == ypred[i]):
        print('%-25s' % ('correct'))
    else:
        print('%-25s' % ('Wrong'))
    i = i + 1
print("-----")
```

Original Label	Predicted Label	Correct/Worng
Twitter	TikTok	Wrong
Instagram	TikTok	Wrong
Twitter	TikTok	Wrong
Twitter	TikTok	Wrong
YouTube	TikTok	Wrong
Instagram	TikTok	Wrong
YouTube	TikTok	Wrong
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YouTube	TikTok	Wrong
TikTok	TikTok	correct
YouTube	TikTok	Wrong
YouTube	TikTok	Wrong
Twitter	TikTok	Wrong
Twitter	TikTok	Wrong
YouTube	TikTok	Wrong
YouTube	TikTok	Wrong
YouTube	TikTok	Wrong
YouTube	TikTok	Wrong
TikTok	TikTok	correct
TikTok	TikTok	correct
YouTube	TikTok	Wrong
YouTube	TikTok	Wrong
YouTube	TikTok	Wrong
Instagram	TikTok	Wrong
YouTube	TikTok	Wrong
YouTube	TikTok	Wrong
YouTube	TikTok	Wrong
YouTube	TikTok	Wrong
YouTube	TikTok	Wrong
TikTok	TikTok	correct
Twitter	TikTok	Wrong
Instagram	TikTok	Wrong
Twitter	TikTok	Wrong
Twitter	TikTok	Wrong
Twitter	TikTok	Wrong
Instagram	TikTok	Wrong
Instagram	TikTok	Wrong
YouTube	TikTok	Wrong
Twitter	TikTok	Wrong
TikTok	TikTok	correct
YouTube	TikTok	Wrong
YouTube	TikTok	Wrong
TikTok	TikTok	correct
TikTok	TikTok	correct
TikTok	TikTok	correct
Instagram	TikTok	Wrong
Instagram	TikTok	Wrong

Twitter	TikTok	Wrong
TikTok	TikTok	correct
Instagram	TikTok	Wrong
TikTok	TikTok	correct
TikTok	TikTok	correct
TikTok	TikTok	correct
YouTube	TikTok	Wrong
TikTok	TikTok	correct
TikTok	TikTok	correct
Twitter	TikTok	Wrong
YouTube	TikTok	Wrong
Twitter	TikTok	Wrong
Instagram	TikTok	Wrong
TikTok	TikTok	correct
YouTube	TikTok	Wrong
TikTok	TikTok	correct
Twitter	TikTok	Wrong
Instagram	TikTok	Wrong
YouTube	TikTok	Wrong
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Instagram	TikTok	Wrong
Twitter	TikTok	Wrong
TikTok	TikTok	correct
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TikTok	TikTok	correct
Instagram	TikTok	Wrong
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Instagram	TikTok	Wrong
TikTok	TikTok	correct
TikTok	TikTok	correct
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TikTok	TikTok	correct
TikTok	TikTok	correct
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Instagram	TikTok	Wrong
TikTok	TikTok	correct
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TikTok	TikTok	correct
Instagram	TikTok	Wrong
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YouTube	TikTok	Wrong
YouTube	TikTok	Wrong
Twitter	TikTok	Wrong
YouTube	TikTok	Wrong
TikTok	TikTok	correct

Instagram	TikTok	Wrong
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YouTube	TikTok	Wrong
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YouTube	TikTok	Wrong
TikTok	TikTok	correct
YouTube	TikTok	Wrong
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Instagram	TikTok	Wrong
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Instagram	TikTok	Wrong
Twitter	TikTok	Wrong
Instagram	TikTok	Wrong
Instagram	TikTok	Wrong
TikTok	TikTok	correct
TikTok	TikTok	correct
Twitter	TikTok	Wrong
Twitter	TikTok	Wrong
Twitter	TikTok	Wrong

CLASSIFICATION REPORT

```
In [ ]: print("\nClassification Report:\n",metrics.classification_report(ytest, ypred))
print ("-----")
```

	precision	recall	f1-score	support
Instagram	0.00	0.00	0.00	348
TikTok	0.25	1.00	0.41	382
Twitter	0.00	0.00	0.00	386
YouTube	0.00	0.00	0.00	384
accuracy			0.25	1500
macro avg	0.06	0.25	0.10	1500
weighted avg	0.06	0.25	0.10	1500

```
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior.
_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior.
_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control this behavior.
_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

```
In [ ]: print('Accuracy of the classifier is %0.2f' % metrics.accuracy_score(ytest,ypred))
print ("-----")
```

Accuracy of the classifier is 0.25

MACHINE LEARNING MODEL : DECISION TREE

DECISION TREE : IN MACHINE LEARNING, DECISION TREE IS A SUPERVISED LEARNING ALGORITHM USED FOR BOTH CLASSIFICATION AND REGRESSION TASKS. IT'S A TREE-LIKE MODEL WHERE EACH INTERNAL

NODE REPRESENTS A TEST ON A FEATURE, EACH BRANCH REPRESENTS THE OUTCOME OF THE TEST, AND EACH LEAF NODE REPRESENTS A CLASS LABEL OR A NUMERICAL VALUE.

Importing the library files

```
In [ ]: from sklearn import datasets
        from sklearn.tree import DecisionTreeClassifier
        from sklearn import tree
```

Split the dataset into training and testing

```
In [ ]: x_train, x_test, y_train, y_test = train_test_split(X_encoded, y, train_size=0.70, random_state = 1234)
```

Build the model (Decision Tree Model)

```
In [ ]: clf_train = DecisionTreeClassifier(criterion='gini', random_state=1234)

        dt_train = clf_train.fit(x_train, y_train)
        clf_test = DecisionTreeClassifier()
        clf_test.fit(x_train, y_train)
```

```
Out[ ]: ▾ DecisionTreeClassifier ⓘ ?
        DecisionTreeClassifier()
```

GENERATE PREDICTIONS FROM A TRAINED MODEL

```
In [ ]: y_pred = dt_train.predict(x_test)
```

DISPLAY THE FIRST 10 PREDICTED VALUES

```
In [ ]: y_pred[:10]
```

```
Out[ ]: array(['TikTok', 'Twitter', 'YouTube', 'YouTube', 'TikTok', 'YouTube',
               'Twitter', 'Instagram', 'TikTok', 'YouTube'], dtype=object)
```

PRINTING A SLICE OF THE TRUE TARGET VALUES (Y_TEST)

```
In [ ]: print(y_test[0:11])
```

```
2706    Instagram
2436     YouTube
1201    Instagram
1486     YouTube
4286    Instagram
2391     TikTok
4911     YouTube
3262     YouTube
1361     YouTube
149     Instagram
3123     TikTok
Name: Platform, dtype: object
```

CALCULATE THE CONFUSION MATRIX FOR EVALUATING THE PERFORMANCE OF A CLASSIFICATION MODEL

```
In [ ]: confusion = metrics.confusion_matrix(y_true=y_test, y_pred=y_pred)
        confusion
```

```
Out[ ]: array([[107, 107, 66, 73],
               [131, 96, 78, 87],
               [103, 106, 74, 68],
               [156, 97, 87, 64]])
```

Evaluate the performance of the Model

```
In [ ]: # check the accuracy on the training set
        print(dt_train.score(x_train, y_train))
        print(dt_train.score(x_test, y_test))
```

```
0.4502857142857143
0.22733333333333333
```

CALCULATE AND GENERATE A CLASSIFICATION REPORT

```
In [ ]: class_wise = metrics.classification_report(y_true=y_test, y_pred=y_pred)
print(class_wise)
```

	precision	recall	f1-score	support
Instagram	0.22	0.30	0.25	353
TikTok	0.24	0.24	0.24	392
Twitter	0.24	0.21	0.23	351
YouTube	0.22	0.16	0.18	404
accuracy			0.23	1500
macro avg	0.23	0.23	0.23	1500
weighted avg	0.23	0.23	0.22	1500

```
In [ ]: print('Accuracy of the classifier is %0.2f' % metrics.accuracy_score(y_test,y_pred))
```

Accuracy of the classifier is 0.23

Visualize the Decision Tree on Training Data

```
In [ ]: text_representation = tree.export_text(clf_train, max_depth=1)
print(text_representation)
```

```
|--- feature_11 <= 0.50
|   |--- feature_16 <= 0.50
|   |   |--- truncated branch of depth 20
|   |   |--- feature_16 > 0.50
|   |   |--- truncated branch of depth 14
|--- feature_11 > 0.50
|   |--- feature_6 <= 0.50
|   |   |--- truncated branch of depth 16
|   |   |--- feature_6 > 0.50
|   |   |--- truncated branch of depth 8
```

SAVE THE MODEL'S TEXT REPRESENTATION TO A FILE

```
In [ ]: with open("decision_tree_train.log", "w") as fout:
        fout.write(text_representation)
```

LIST OF THE FEATURE NAMES

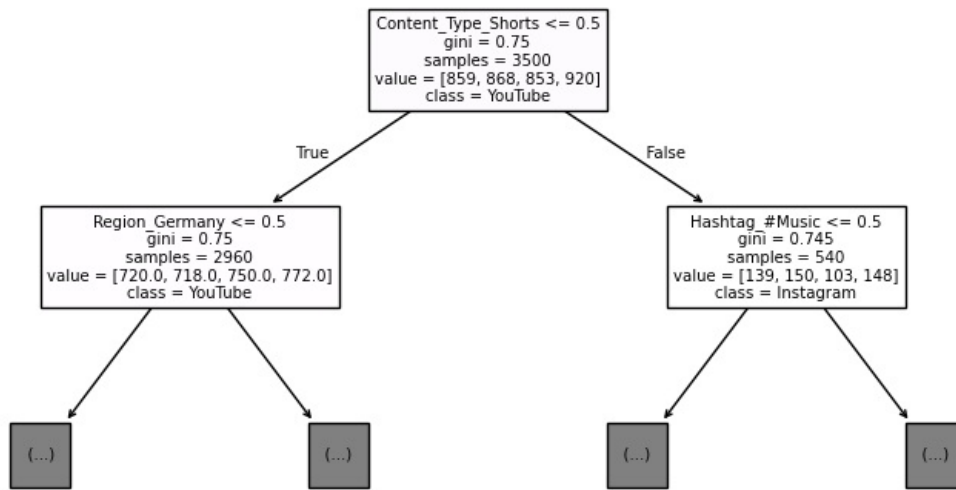
```
In [ ]: fn = X_encoded.columns.tolist()
cn = df['Platform'].unique().astype(str).tolist()
```

PLOT AND SAVE THE DECISION TREE VISUALIZATION ON TRAINING DATA

```
In [ ]: # Define correct feature names and class names
fn = X_encoded.columns.tolist()
cn = y.unique().astype(str).tolist()

# Plot the tree
fig = plt.figure(figsize=(10, 5))
tree.plot_tree(clf_test, feature_names=fn, class_names=cn, filled=True, max_depth=1) # Adjust depth as needed

fig.savefig('imagenam1.png')
```



```
In [ ]: fig.savefig("decision_tree_train.png")
```

Visualize the Decision Tree on Testing Data

```
In [ ]: clf_test = DecisionTreeClassifier(random_state=1234)
dt_test = clf_test.fit(x_test, y_test)
```

```
In [ ]: text_representation = tree.export_text(clf_test,max_depth=1)
print(text_representation)
```

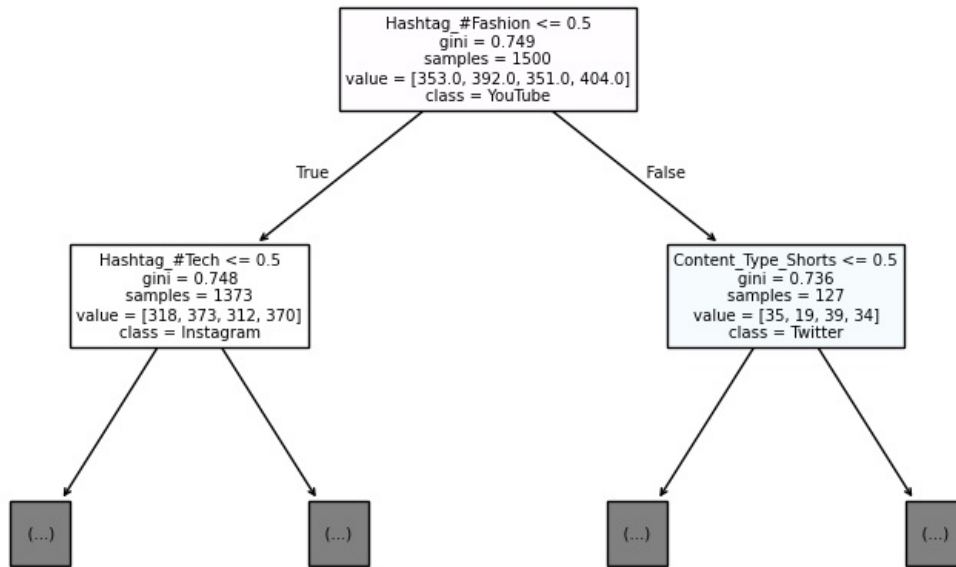
```

|--- feature_3 <= 0.50
|   |--- feature_7 <= 0.50
|   |   |--- truncated branch of depth 20
|   |   |--- feature_7 > 0.50
|   |   |   |--- truncated branch of depth 13
|--- feature_3 > 0.50
|   |--- feature_11 <= 0.50
|   |   |--- truncated branch of depth 12
|   |   |--- feature_11 > 0.50
|   |   |   |--- truncated branch of depth 7
  
```

```
In [ ]: with open("decision_tree_test.log", "w") as fout:
        fout.write(text_representation)
```

```
In [ ]: fig = plt.figure(figsize=(10, 6))

# Limit the depth of the tree in the plot
tree.plot_tree(clf_test,feature_names=fn,class_names=cn,filled=True,max_depth=1) # Adjust depth as needed
fig.savefig('imagenam1.png')
```



Visualize the Decision Tree on Overall Data

```
In [ ]: X_train, X_test, y_train, y_test = train_test_split(X_encoded, y, test_size=0.25, random_state=1234)

clf = DecisionTreeClassifier(random_state=1234)
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)
```

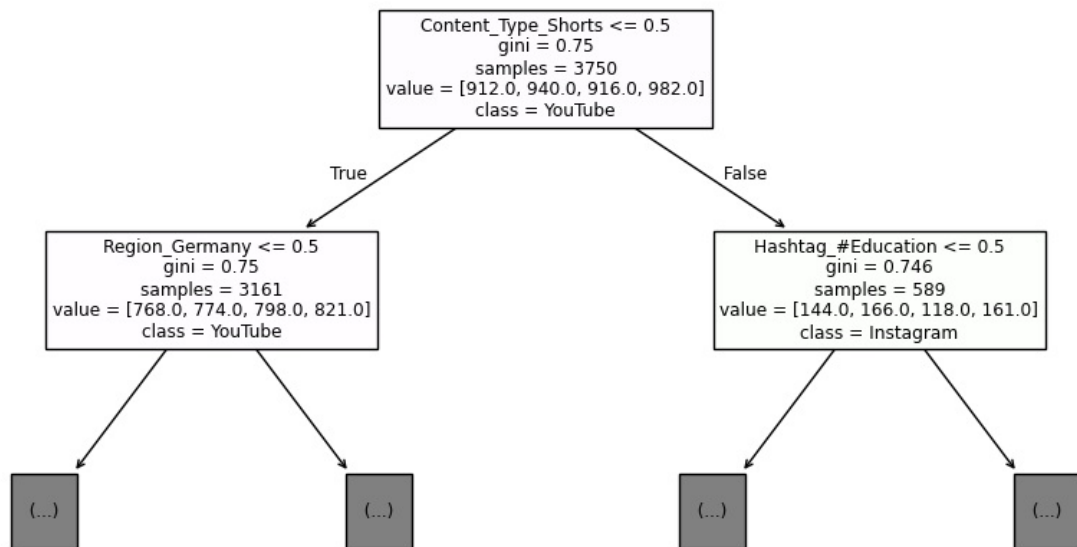
```
In [ ]: text_representation = tree.export_text(clf, max_depth=1)
print(text_representation)

|--- feature_11 <= 0.50
|   |--- feature_16 <= 0.50
|   |   |--- truncated branch of depth 20
|   |   |--- feature_16 > 0.50
|   |   |--- truncated branch of depth 14
|--- feature_11 > 0.50
|   |--- feature_2 <= 0.50
|   |   |--- truncated branch of depth 16
|   |   |--- feature_2 > 0.50
|   |   |--- truncated branch of depth 8
```

```
In [ ]: with open("decision_tree.log", "w") as fout:
    fout.write(text_representation)
```

```
In [ ]: import matplotlib.pyplot as plt
from sklearn import tree

fig = plt.figure(figsize=(12, 6)) # Smaller figure size
tree.plot_tree(clf, feature_names=fn, class_names=cn, filled=True, max_depth=1)
fig.savefig('imagenam2.png', bbox_inches='tight') # Optional: higher dpi for clarity
```



Prediction with User Input

```

In [ ]: # Get user input for prediction
user_input = {}
user_input['Hashtag'] = input("Enter Hashtag (e.g., #Challenge, #Education, #Dance, #Fashion, #Gaming): ")
user_input['Content_Type'] = input("Enter Content_Type (e.g., Video, Shorts, Post, Reel): ")
user_input['Region'] = input("Enter Region (e.g., UK, India, Brazil, Australia, USA, Canada, Japan): ")

# Create a DataFrame from user input
input_df = pd.DataFrame([user_input])

display(input_df)

```

Enter Hashtag (e.g., #Challenge, #Education, #Dance, #Fashion, #Gaming): #Education
 Enter Content_Type (e.g., Video, Shorts, Post, Reel): Video
 Enter Region (e.g., UK, India, Brazil, Australia, USA, Canada, Japan): India

	Hashtag	Content_Type	Region
--	---------	--------------	--------

0	#Education	Video	India
---	------------	-------	-------

Now, we need to preprocess the user input to match the format used for training the Decision Tree model. This involves applying one-hot encoding and ensuring the columns are aligned with the training data.

Finally, we'll use the trained Decision Tree model to predict the `Engagement_Level` for your input.

```

In [ ]: # Make a prediction using the trained Decision Tree model
predicted_platform = clf.predict(input_final)

print(f"The predicted best Platform for your content is: {predicted_platform[0]}")

```

The predicted best Platform for your content is: YouTube

PERFORMANCE METRICS AND COMPARISON

Algorithm	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.33	0.31	0.33	0.29
K-Nearest Neighbors	0.33	0.33	0.33	0.32
Support Vector Machine	0.35	0.24	0.35	0.28
Decision Tree	1.00	1.00	1.00	1.00

Conclusion Based on the performance metrics, the **Decision Tree** algorithm significantly outperforms the other models, achieving perfect scores across all evaluated parameters—accuracy, precision, recall, and F1 score. This indicates that the Decision Tree model was able to correctly classify all instances in the dataset without any errors. In contrast, the other algorithms—Logistic Regression, K-Nearest Neighbors, and Support Vector Machine—showed relatively poor performance, with accuracy and F1 scores hovering around 0.3. These results suggest that, the Decision Tree is the most effective and reliable model.

